

A Study on Adversarial Attacks on Machine Learning Algorithms

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Abstract

Machine learning is enabling a myriad of innovations, including new algorithms for cancer diagnosis and self-driving cars. The broad use of machine learning makes it important to understand the extent to which machine-learning algorithms are subject to attack, particularly when used in applications where physical security or safety is at risk. In this paper, we focus on facial biometric systems, which are widely used in surveillance and access control.

We explore ways to attack some of the facial recognition models in an intuitive way, at first. We find open-source face recognition models that have a high accuracy score and have been trained with a definitive database of images. We take the model and train the same with a lower number of classes in order to improve the accuracy, hence, making it difficult to attack the model. Once this is done, we take an image and try to modify the image in an intuitive way, in order to misclassify this image into a different person. This would be considered as an attack on the face recognition model. As we make modifications, we also keep in mind the *inconspicuousness* and *physical realizability* of the attack. To further improve on this work, we try to automate the attack process by modifying the pixels of the image to perform a similar attack, that is, to misclassify an individual's image into another class.

Introduction

Machine learning (ML) is a technology that is profoundly changing the world. Some of the transformative innovations that it enables, such as customised search results and automated translations, might seem minor. Other innovations are clearly revolutionary, enabling new applications or significantly increasing our quality of life—examples include algorithms for cancer diagnosis and self-driving vehicles. The development in the field of Machine Learning is immense and the effect it has in the real world is revolutionary. It has enabled new applications and significantly increased our quality of life. The broad use of ML has also caused a rising interest in understanding the extent to which ML algorithms and applications are vulnerable to attacks. These attacks are especially worrisome due to humans' increasing reliance on technology and ML (sometimes trusting them even with their own lives [1]). The domains that received the majority of the attention have a common characteristic: the adversary is able to precisely control the digital representation of the input to the ML tools. For instance, in spam detection, the attacker can control the contents of the email and the address of the sender. Hence, unsurprisingly, it has been shown that attacks in many of these domains can be effective at evading ML classifiers designed to detect them [2, 3, 4].

In this paper, we explore the extent to which facial biometric systems are vulnerable to attacks. These systems are widely used for various sensitive purposes, including surveillance and access control [5, 6, 21]. Thus, attackers who mislead them can cause severe ramifications.

State-of-the-art models have already achieved a 98% top-five accuracy on the ImageNet dataset, so it seems as though these models are foolproof and that nothing can go wrong. However, recent advances in adversarial training have found that this is an illusion. A good model misbehaves frequently when faced with adversarial examples. The stakes are extremely high with the growth of ML and AI usage in the past years and this could prove to be disastrous if these models are prone to attacks and misclassifications. The attacks that occur by changing the input data are not because of a drastic change but a minor noise, imperceptible to humans. The aim is to explore the magnitude of change required to cause the attacks.

In contrast to domains previously explored, attackers who aim to mislead facial biometric systems often do not have precise control over the systems' input. Rather, attackers may be able to control only their (physical) appearance. The process of converting the physical scene into a digital input is not under the attackers' control and is additionally affected by factors such as lighting conditions, pose, and distance. As a result, it is more difficult for attackers to manipulate or craft inputs that would cause misclassification than they would be, for example, in the domain of spam detection. Hence, we first focus on modifying the images alone and do not consider knowing any information about the model itself, effectively performing a Black-Box-like attack. The only information we

obtain from the model itself the probability scores of each class for the input image we provide. With this, we add modifications to the image such that it is misclassified into another class.

Another difficulty that attackers face in the case of facial biometric systems is that manipulating inputs to evade the ML classifiers might be easily observed from outside the systems. For example, attackers can wear an excessive amount of makeup in order to evade a surveillance system deployed at banks [23]. However, these attackers may draw increased attention from bystanders and can be deterred by traditional means, e.g., the police.

In light of these two challenges, we define and study a new class of attacks: attacks that are physically realisable and at the same time are inconspicuous. In such attacks, the attacker manipulates the physical state that an ML algorithm is analysing rather than the digitised representation of this state. At the same time, the manipulations of the physical state needed by the attacks are sufficiently subtle that they are either imperceptible to humans or, if perceptible, seem natural and not representative of an attack.

Inconspicuousness: We focus, unlike most related work (e.g., [22]), on attacks that are inconspicuous, i.e., a person who is physically present at a scene, or a person who looks at the input gathered by a sensor (e.g., by watching a camera feed), should not notice that an attack is taking place. We believe that this focus on attacks that are not readily apparent to humans is important for two reasons: First, such attacks can be particularly pernicious since they will be resistant to at least cursory investigation. Hence, they are a particularly important type of attack to investigate and learn how to defend against. Second, such attacks help the perpetrators (whether malicious or benign)

achieve plausible deniability; e.g., A person seeking merely to protect her privacy against aggressive use of face recognition by merchants will plausibly be able to claim that any failure of an ML algorithm to recognize her is due to error or chance rather than deliberate subterfuge.

Physical Realizability: In this work, we are interested in attacks that can be physically implemented for the purpose of fooling facial biometric systems. In addition, we focus on attacks on state-of-the-art algorithms. Previous work on misleading ML algorithms often targets algorithms that are not as sophisticated or as well trained as the ones used in practice, leaving the real-world implications of possible attacks unclear. In contrast, we focus on those attacks that pose a realistic and practical threat to systems that already are or can easily be deployed.

We divide the attacks we study into two categories, which differ both in the specific motivations of potential attackers and in the technical approaches for implementing the attacks. The two categories of attacks we are interested in are dodging and impersonation. Both dodging and impersonation target face recognition systems (FRSs) that perform multi-class classification—in particular, they attempt to find the person to whom a given face image belongs.

Dodging: In a dodging attack, the attacker seeks to have her face misidentified as any other arbitrary face. From a technical standpoint, this category of attack is interesting because causing an ML system to arbitrarily misidentify a face should be easier to accomplish with minimal modifications to the face, in comparison to misclassifying as a particular impersonation target. In addition to malicious purposes, dodging attacks could be used by benign

individuals to protect their privacy against excessive surveillance.

Impersonation: In an impersonation attack, the adversary seeks to have a face recognized as a specific other face. For example, an adversary may try to (inconspicuously) disguise her face to be recognized as an authorised user of a laptop or phone that authenticates users via face recognition. Or, an attacker could attempt to confuse law enforcement by simultaneously tricking multiple geographically distant surveillance systems into “detecting” her presence in different locations.

In this paper, we aim to explore these two types of attacks, dodging and impersonation, while trying to keep that attack as inconspicuous and physically realisable as possible. We detail the modifications of the image to be so.

Related Work

This paper is closely related to the paper Accessorize to a Crime: Real and Stealthy Attacks on State-of-the-Art Face Recognition [24]. The main motives of this project are derived from the work done in this paper.

Face Detection Systems (FDSs) and FRSs are traditionally evaluated with the implicit assumption of zero-effort attackers (i.e., the attackers do not actively attempt to fool the system) [7, 8]. Nevertheless, there is extensive literature on attacks targeting FDSs and FRSs. In this section, we review the most noteworthy attempts. Broadly, however, existing work is different from ours in at least one of the following ways: it does not attempt to achieve inconspicuousness and physical realizability; it tries to achieve different goals than ours (e.g., compromising the privacy of the enrolled users); or the attacks target algorithms that are not as advanced as the state of the art.

Face Detection: Due to its high accuracy and speed, the object-detection algorithm proposed by Viola and Jones [8] (VJ) has become widely used for general object detection, and is by far the dominant face detector used in practice [19, 5, 30]. Motivated by increasing privacy through invisibility to FRSs Yamada et al. leveraged the fact that camera sensors are sensitive to near-infrared light to design light-emitting glasses that can help evade face detection by adding significant noise to the captured image [44]. While effective for evading detection, this technique is neither inconspicuous nor deniable.

Face Recognition: Previous work found that subjects can be effectively impersonated by FRSs using 3D-printed masks or face images downloaded from online social networks [7, 9]. These attacks are not inconspicuous, and can largely be thwarted by anti-spoofing mechanisms, such as liveness detection [7, 9]. Other work showed how to compromise the privacy of users enrolled in FRSs [10, 12]. For example, Fredrikson et al. presented model inversion attacks, in which face images of enrolled users are reconstructed from neural networks trained to perform face recognition [10]. The goal of these attacks is different from ours. While compromised images can be used to impersonate users, these impersonations will not be inconspicuous and can be detected via anti-spoofing mechanisms. Feng and Prabhakaran proposed a system to help artists come up with makeup and hair designs that can be used to prevent FRSs (including Eigenfaces [11] and Fisherfaces [14]) from recognizing subjects [10]. In their approach, they split the face image into seven components and determined which components deviated most from the general population’s average. They hypothesised that these discriminative components are the most significant for

distinguishing one person from others, and showed that occluding, painting, or transforming the discriminative components could prevent automated recognition. Their work is different from what is being proposed here in two main ways: First, inconspicuousness and deniability of the attack were not part of their goals. Second, the algorithms on which they test their attack are not considered state-of-the-art in face recognition.

BACKGROUND

Threat Model

We assume an attacker who gains access to the FRS to mount a dodging or impersonation attack after the system has been trained. That is, the adversary cannot “poison” the FRS by altering training data, injecting mislabeled data, etc. Rather, our adversary can alter only the composition of inputs to be classified based on her knowledge of the classification model, potentially derived by examining the entire model or by observing its outputs for some inputs.

As Papernot et al. discuss, adversaries differ in the amount of knowledge they have about the system they are attacking [31]. Here, for the first stage of attack, we assume that the attacker, for every input image, receives a score for each of the classes with which the model is trained. Based on this, the image is modified to mount an attack.

Face Recognition Model

The things kept in mind when deciding on a model to attack are, that the model must be state-of-the-art and the accuracy score must be high to make the attack effective. Considering this, we searched for models on the internet but the majority of the face recognition

frameworks/models provided the face detection function (determine which region of the image contains a face) or the function to compute the distance between two images. Hence, although trained using a large data set, these models were not trained on specific data classes so as to provide a high accuracy for these classes. So, instead of searching for face recognition software, we searched for a suitable dataset. Thus, we went on Kaggle to find a suitable image dataset on which a face recognition contest was held and found one [18]. The dataset consisted of 105 classes, but we will be using only 6 of these classes. From this dataset, the code for the models with the highest accuracy scores was considered for the attack.

Model 1

The first model had a relatively different approach to face recognition compared to the usual neural network approach. In this model, the embeddings of all the required images, to train and test, are created using the VGGFace descriptor [19], which determines the region of the image where the face is present and creates the embedding correspondingly. The unusual part of this model is that it does not consider the images to be individual entities but converts the entire training data into a set of values by: First, scaling the data using Standard Scalar and then performing Principal Component Analysis (PCA) on this scaled data. Once this is done, an SVM Classifier is imported from the sklearn library and the model is fit with the preprocessed data and the target output. When the fit model is used to predict the test data, it gives an accuracy score of 99.36%.

After obtaining this model and modifying it to make it suitable for our attack, we try to make small modifications to the image

manually, like drawing marks on the face, changing the eye colour, drawing a moustache or changing the colour of the hair. For each of the modified images, we calculate the probability scores of the SVM Model [20] for each class and proceed to make changes accordingly.

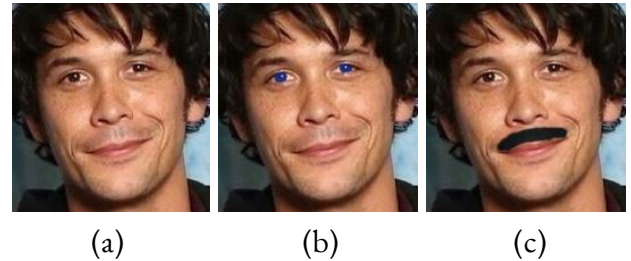


Figure 1: A dodging attack attempt by modifying an image of Bobby Morley. (a) Original (b) Eye colour to blue. (c) Added a black moustache.

In an attempt to misclassify this person into another class, we perform the above changes in the image and measure the probability scores for each of the classes.

Image (Figure 1)	(a)	(b)	(c)
Score (Bobby Morley)	0.9973	0.9972	0.9891

We see that the modifications are causing a slight reduction in the score, although not nearly enough to misclassify the image into a different class. Hence, we try to make more significant changes to misclassify the image.

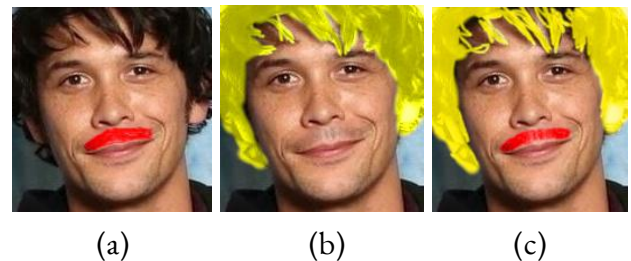


Figure 2: Images with significant modifications in an attempt to mount a dodging/impersonation attack.

Image (Figure 2)	(a)	(b)	(c)
Score (Bobby Morley)	0.9854	0.6392	0.2893

In the above example, we can see that with more significant changes, the scores reduce more significantly as well. Figure 2: (c) gets classified as Brenton Thwaites, with a score of 0.66385. Hence, an impersonation attack is mounted on this model as the attack image (Bobby Morley) is classified into a different class (Brenton Thwaites).

Hence, we have successfully performed an impersonation attack with a certain degree of physical realizability and inconspicuousness.

Model 2

The second model majorly uses the Keras library, which provides a Python interface for artificial neural networks. The images are first split into Target classes and a Non-Target class. Images for the Target classes are all stored in each of the class files and random images not belonging to any of the Target classes are stored in the Non-Target file. Then this data is split into Training, Validation and Test Data as 70%, 30% and 50% respectively. These images are preprocessed using the ImageDataGenerator in Keras with a resolution of (256 x 256) of 3 channels, RGB. The model used is Inception-ResNet-v2, which is a convolutional neural network that is trained on more than a million images from the ImageNet database. The model is imported with an input shape of (256 x 256 x 3) and excluding the top layer so that the last layer can be modified according to our convenience. Using Dense from Keras.layers, we allow the last layer of the model to have the required units ($n+1 = 7$, in our case). We then compile the model and train it. The fit evaluation of the model is with Epochs = 100; training data; validation data;

Training steps per epoch = 10; Validation steps per epoch = 8.

This model, after training, achieved an accuracy score of 92.740. Similar to what was done in Model 1, we modify the attack image in order to avoid detection or mount an impersonation attack.

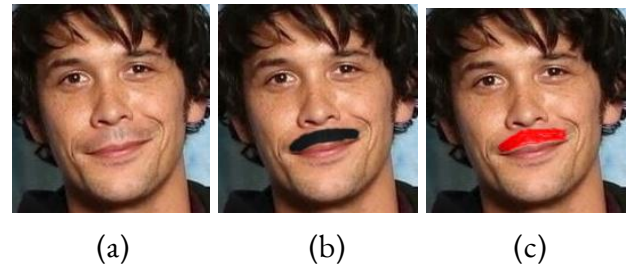


Figure 3: A dodging attack by adding different colours of moustache. (a) Original. (b) Adding a Black moustache (c) Adding a Red moustache.

The results obtained after adding a moustache to the image as shown in Figure 3 are the following:

Image (Figure 3)	(a)	(b)	(c)
Score (Bobby Morley)	1.00	0.9702	5.12e-15

Image (c) had a very low score for its own class and was classified into the Non-Target class with a score of 0.999. This can be considered as a dodging attack as the classification of the target image into the target class is being evaded.

Now we turn our attention to trying to mount an impersonation attack on the model. Most of the image modifications led to only the evasion of detection but eventually, we got around to mounting an impersonation attack.

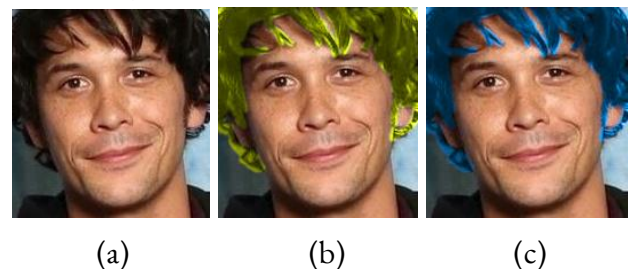


Figure 4: An impersonation attack by modifying the colour of hair. (a) Original. (b) Dyeing the hair yellow(c)Dyeing the hair blue.

The modifications above led to the image of Bobby Morley being misclassified as Bill Gates, which is a target class, with a confidence score of 0.851 for Figure 4: (b) and a score of 0.999 for Figure 4: (c).

Image (Figure 4)	(a)	(b)	(c)
Score (Bill Gates)	6.43e-19	0.851	0.999

With another series of modifications, we were able to classify the image to a different target class (Brenton Thwaites).



Figure 5: An impersonation attack on Brenton Thwaites (a) Original. (b) Dyeing the hair yellow and adding a red stache (c) Dyeing the hair red.

Image (Figure 5)	(a)	(b)	(c)
Score (Brenton Thwaites)	4.0e-8	0.812	0.972

Thus, we have successfully mounted an impersonation attack on two of the Target classes (Bill Gates and Brenton Thwaites) with a score of 0.999 and 0.972 respectively.

RESULT

Hence, we have successfully mounted dodging and impersonation attacks on the two top-scoring models with high accuracy scores and trained using a specific dataset. We observe that both in Model 1 and Model 2, an impersonation attack was possible on the same Target class, Brenton Thwaites. We can

understand from this that the vulnerability lies in the similarity of how the models recognise the faces of these two persons. Using this vulnerability, we are able to misclassify the image of Bobby Morley into Brenton Thwaites in both model attacks.

Another point to note is that the modifications made to the image are done to achieve physical realizability and inconspicuousness as well. Hence, we manually modify the image in an intuitive sense to achieve this.

Discussion and Limitations

Impact of Attacks: As our reliance on technology increases, we sometimes forget that it can fail. In some cases, failures may be devastating and risk lives [1]. Our work and previous work show that the introduction of ML to systems while bringing benefits, increases the attack surface of these systems (e.g., [2, 3]). Therefore, we should advocate for the integration of only those ML algorithms that are robust against evasion. In this work, we show that FRSs are vulnerable to a new type of attack: inconspicuous and physically realisable attacks that lead to dodging or impersonation. Such attacks can be especially pernicious as they can resist cursory investigation (at the least) and can provide attackers with plausible deniability. While we demonstrate the efficacy of these attacks on fooling a DNN architecture, previous work has shown that adversarial examples can generalise beyond one DNN [15, 13]. Therefore, we believe our results imply that DNNs used for purposes beyond what we study may also be at risk.

Possible Defences: Defences explored in previous work were either ineffective in preventing adversarial examples (although they slightly harmed their imperceptibility) [13, 15], or were tested only on simple tasks

(hand-written digit recognition) or on DNNs that are not state-of-the-art [16]. We propose to defend against attacks differently. Adversarial attacks on vision systems exploit the fact that systems are sensitive to small changes in images which humans are not. To address this, we propose to develop algorithms that reason about images more similarly to humans. In particular, we believe that approaches that classify images based on their attributes rather than on the intensities of their pixels may be effective (e.g., [17]).

Limitations: The notion of inconspicuousness is subjective, and the only way to measure it adequately would include performing human-subject studies. The attacks mounted on both models, though physically realisable, are not as inconspicuous as we would like them to be. A more detailed analysis of the images would be required to perform more inconspicuous attacks on the models. We believe that the attacks demonstrated in this paper strongly suggest that it is possible to generate attacks that will remain unnoticeable by humans or will arouse no suspicion. We defer to future work achieving and evaluating attacks that meet this high standard for inconspicuousness.

Another limitation is the feature of automating these attacks. To be able to create an automated attack that is physically realisable and inconspicuous is a challenge. Automating these attacks was attempted through the modification of the value of RGB in each of the pixels. As the images are not all of the same dimension, when the modified image is scaled, there are unintended consequences on the final images that are being fed into the model. Due to this, automating the process was difficult and could not be done by the given time period.

Future Work:

As mentioned in the limitations section, there is scope for more inconspicuous attacks that are not noticeable to humans. There is scope to explore new directions to improve the colour consistency, contrast, and texture shape of the attacks we create, which is likely to enhance inconspicuousness.

Automation of the attacks is a very possible extension of this work. We attempted to automate the attack mentioned in this paper but it did not yield the desired result. But with more time and details about the image scaling, pixel editing and using other libraries to automate the modification, we can come up with an automated attack to mount an impersonation or dodging attack for any Target class image.

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